Compactness for the tableau calculus

The Open Logic Project contributors

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Editorial scope. Statement and complete author proof are included verbatim as the span lines 752-778 of the pinned original source; no text is rewritten or paraphrased.

Allowed standard prerequisites (editorial).

- the tableau derivation relation Gamma proves A as defined in this unit
- consistency of a set of sentences defined by absence of a closed tableau
- tableaus grow from a finite initial list of signed formulas

Proposition 0.1 (Compactness)**.** *1. If* $\Gamma \vdash !A$ *then there is a finite subset* $\Gamma_0 \subseteq \Gamma$ *such that* $\Gamma_0 \vdash !A$*.*

2. If every finite subset of Γ *is consistent, then* Γ *is consistent.*

Proof. 1. If $\Gamma \vdash !A$, then there is a finite subset $\Gamma_0 = \{!B_1, \ldots, !B_n\}$ and a closed !!tableau for

$$\{\bot(!A), \top(!B_1), \ldots, \top(!B_n)\}$$

This !!tableau also shows $\Gamma_0 \vdash !A$.

2. If Γ is inconsistent, then for some finite subset $\Gamma_0 = \{!B_1, \ldots, !B_n\}$ there is a closed !!tableau for

$$\{\top(!B_1), \ldots, \top(!B_n)\}$$

This closed !!tableau shows that Γ_0 is inconsistent.

□

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